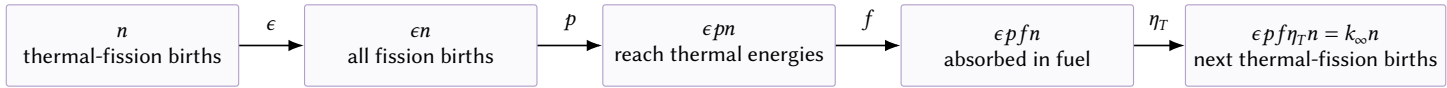


Thermal cells and the four factors

Which part of the neutron life cycle explains a change in multiplication?

17
LESSON

Read: § 4.4 and the four-factor development in § 4.5; Eqs. (4.17)–(4.28), (4.42)–(4.51).



Idealized no-leakage cycle (adapted from Fig. 4.5). Thermal fission produces *fast* neutrons.

1 Keep space and energy together

At thermal and resonance energies, larger optical thicknesses produce substantial fuel/moderator flux differences. Use **regional volume averages** $\phi^r(E)$; the flux need not actually be uniform inside a homogeneous material region.

Groups: T : $E < 1$ eV; I : 1 eV $\leq E < 0.1$ MeV; F : $E \geq 0.1$ MeV.

“Nonthermal” means $I \cup F$, not just F .

For group g and region r , introduce volume-integrated absorption and fission-neutron production:

$$A_g^r = V^r \int_g \Sigma_a^r(E) \phi^r(E) dE,$$

$$B_g^r = V^r \int_g \nu \Sigma_f^r(E) \phi^r(E) dE.$$

Only fuel fissions: $B_g = B_g^f$, $B = \sum_g B_g$, and $A = \sum_{g,r} A_g^r$. With $A_T = A_T^f + A_T^m$, the chapter’s two-region model retains

$$B \simeq B_T + B_F, \quad A \simeq A_T + A_I^f,$$

and hence $k_\infty \simeq (B_T + B_F)/(A_T + A_I^f)$. Eqs. (4.18)–(4.23): intermediate fission, fast absorption, and nonthermal moderator absorption are omitted.

2 Homogenize reaction rates, not atoms alone

With $\bar{\phi}_g^r = \int_g \phi^r(E) dE$ and region-specific group data,

$$\bar{\Sigma}_{xg}^{\text{hom}} = \frac{\sum_r V^r \bar{\Sigma}_{xg}^r \bar{\phi}_g^r}{\sum_r V^r \bar{\phi}_g^r}.$$

Simple volume weighting is recovered only when the regional group fluxes are equal. Material mixing before transport and reaction-rate homogenization after transport are different operations.

3 Read each ratio in plain English

$$\epsilon = \frac{B_T + B_F}{B_T} \quad \text{fast-fission factor,}$$

$$p = \frac{A_T}{A_T + A_I^f} \quad \text{resonance escape,}$$

$$f = \frac{A_T^f}{A_T} \quad \text{thermal utilization,}$$

$$\eta_T = \frac{B_T}{A_T^f} \quad \text{reproduction factor.}$$

ϵ counts the additional fission-neutron production beyond thermal fission. p is the fraction surviving the model’s resonance losses. f assigns thermal absorptions to fuel; η_T counts fission neutrons *per fuel absorption*, not per fission and not per fissile absorption alone.

$$k_\infty = \epsilon p f \eta_T \quad \text{within the retained-term model.}$$

Check the cancellation: the four rate ratios must recover Eq. (4.23), not a different balance.

4 The thermal disadvantage

$$\zeta = \frac{\bar{\phi}_{mT}}{\bar{\phi}_{fT}}, \quad f = \left[1 + \zeta \frac{V_m \bar{\Sigma}_{aT}^m}{V_f \bar{\Sigma}_{aT}^f} \right]^{-1}.$$

A larger moderator-to-fuel thermal flux ratio makes moderator absorption more costly. Include cladding, dissolved boron, and other nonfuel absorbers in a more detailed balance.

OpenMC | Measure the factors before interpreting them

Use CellFilter and an incident-energy EnergyFilter; score flux, absorption, and nu-fission. Reaction-rate tallies already include volume. Divide a flux tally by $V^r \Delta E$ for a spectrum per eV, or by $V^r \Delta u$ for one per lethargy. Compare the chapter’s product, the energy-independent B/A , and the eigenvalue; discuss omitted reactions and statistical uncertainty rather than forcing agreement.

Sources: Ch. 4, §§ 4.4–4.5; slide sets 17–18. OpenMC: Tallies, §§ 8.1–8.2. $B/A = k_\infty$ additionally assumes no leakage or net nonfission neutron production.